

SoS 2026: Algorithmic Trading

Plan of Action

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Timeline

- Week 1** Orientation & First Data: what quant and algorithmic trading actually is and who does it. Study just enough market vocabulary for a coder: instruments, exchanges, returns, volatility; pull real historical price data in Python, clean it, and plot it. (*Chan Ch. 1*)
- Week 2** Fishing for Ideas: where strategies come from, plausible strategies and their pitfalls, the intuition behind mean reversion vs. momentum. Hand-code the signal logic for a simple moving-average crossover and run it on the data from Week 1. (*Chan Ch. 2*)
- Week 3** Backtesting I: Backtesting platforms and data sources, writing my own simple backtester in Python, key performance metrics (Sharpe ratio, maximum drawdown, annualised returns); get a full end-to-end backtest running on my Week-2 strategy. (*Chan Ch. 3*)
- Week 4** Backtesting II: Biases that make backtests lie look-ahead, survivorship, data-snooping plus transaction costs and slippage; deliberately expose and then fix these flaws in my Week-3 backtest; re-evaluate performance honestly. (*Chan Ch. 3*)

Mid-term Report Submission

Deliverable: a working, honestly-evaluated backtested strategy

- Week 5** Mean Reversion & Pairs Trading: build a second, more interesting strategy end-to-end, cointegration testing, spread construction, entry/exit signals, backtest and compare against Week-2 strategy; bridges Chan's practice into Cartea's statistical-arbitrage framing. (*Chan Ch. 2-3; Cartea pairs-trading chapter*)
- Week 6** Market Microstructure & the Order Book: Study electronic market structure, the limit order book (LOB), order types (market, limit, cancel), bid-ask spread, price formation, adverse selection. (*Cartea Ch. 1-2*)
- Week 7** Optimal Execution & Liquidation: price impact (temporary vs. permanent), the optimal liquidation problem, Almgren-Chriss-style execution trajectories, trade scheduling to minimise market impact; the mathematics here directly answers the slippage problems encountered in Week 4. (*Cartea Ch. 5-8*)
- Week 8** Market Making, Risk & Wrap-up: inventory-based and adverse-selection market-making models; money management, position sizing, the Kelly criterion, leverage; assemble the full pipeline. (*Cartea Ch. 10; Chan Ch. 6-8*)

End-term Report Submission

Deliverable: full pipeline + write-up of what you built and learned

References

Algorithmic and High-Frequency Trading by Álvaro Cartea, Sebastián Jaimungal & José Penalva

Quantitative Trading: How to Build Your Own Algorithmic Trading Business (2nd ed.)
by Ernest P. Chan